

MEA OCPP 1.6 Compliance Test Report

Full Summary — All Sections (1–11)

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-18

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1 Test Configuration

Device Under Test Vector vSECC.single Board
Device IP 192.168.1.166
OCPP Version 1.6 (JSON)
CP ID rddQC4000001
CSMS wss://ocpp.measandbox.com:2930
Connection Direct TLS (Security Profile 1, no proxy)
Report Generated 2026-06-18

Test Architecture

The vSECC connects directly to the MEA CSMS over TLS with no intermediate proxy. OCPP traffic is observed via GET /api/logging/files/ocpplib.log (TRACE-level, raw » sent / « received frames) and via MQTT at 192.168.1.166:1883 (vsecc/#). The test PC NAT-routes internet access for the vSECC subnet (192.168.1.x via enp3s0).

2 Overall Summary

Grand total: 66 PASS 0 FAIL 151 WARN 50 SKIP (267 items across all sections)

Sec	Section Title	PASS	FAIL	WARN	SKIP	Total	Date
1	Charger Configuration Verification	14	0	10	0	24	2026-06-18
2	Auto Charge Verification	3	0	18	0	21	2026-06-11
3	Normal Charging Verification	3	0	16	0	19	2026-06-11
4	Reset Verification	6	0	15	0	21	2026-06-11
5	Reservation Verification	6	0	13	0	19	2026-06-11
6	Charging Profile Verification	4	0	22	0	26	2026-06-11
7	Abnormal Operation Verification	9	0	45	0	54	2026-06-11
8	Dual Connector Verification	0	0	0	50	50	2026-06-11
9	MEA-Specific Configuration & V2G/BPT	5	0	3	0	8	2026-06-11
10	CSMS Command Verification	14	0	9	0	23	2026-06-11
11	Performance – Reconnect Time	2	0	0	0	2	2026-06-11
All	Grand Total	66	0	151	50	267	

3 Section 1: Charger Configuration Verification

14 PASS 0 FAIL 10 WARN 0 SKIP (24 total)

Item	Test Description	Detail / Evidence	Result
1.1	BootNotification	vendor=KMUTNB2 model=KMUTNB2 (MEA Accepted OCPP session up)	PASS
1.2	StatusNotification (boot, all connectors)	Not observed on MQTT or ocpplib.log (vSECC likely connected before test started) <i>vendorId/vendorErrorCode absent (optional per OCPP 1.6 4.7)*</i>	WARN
1.3	TriggerMessage(BootNotification) Accepted	MEA sandbox /remote/triggerMessage not exposed (HTTP 404) <i>CSMSCS command; not testable via MEA REST API</i>	WARN
1.4	BootNotification (triggered) fields	Depends on 1.3 (TriggerMessage not available via MEA REST API)	WARN
1.5	TriggerMessage(StatusNotification) Accepted	MEA sandbox /remote/triggerMessage not exposed (HTTP 404) <i>CSMSCS command; not testable via MEA REST API</i>	WARN
1.6	StatusNotification (triggered) fields	Depends on 1.5 (TriggerMessage not available via MEA REST API)	WARN
1.7	TriggerMessage(MeterValues) Accepted	MEA sandbox /remote/triggerMessage not exposed (HTTP 404) <i>CSMSCS command; not testable via MEA REST API</i>	WARN
1.8	MeterValues fields & measurands	Requires active charging session <i>Retest during a charging session</i>	WARN
1.9	GetConfiguration (required configurationKey)	GetConfiguration sent; response not observed in ocpplib.log	WARN
1.10	ChangeConfiguration HeartbeatInterval=600 Accepted	HTTP 200	PASS
1.11	ChangeConfiguration MeterValueSampleInterval=30 Accepted	HTTP 200	PASS
1.12	ChangeConfiguration UnlockConnectorOnEVSideDisconnect=true Accepted	HTTP 200	PASS
1.13	ChangeAvailability(cid=1, Inoperative) Accepted	StatusNotification Unavailable confirmed via ocpplib.log	PASS
1.14	StatusNotification (Inoperative) fields	Unavailable confirmed via MQTT/ocpplib.log <i>vendorId/vendorErrorCode absent (optional per OCPP 1.6 4.7)*</i>	PASS
1.15	ChangeAvailability(cid=1, Operative) Accepted	StatusNotification Available confirmed via ocpplib.log	PASS
1.16	StatusNotification (Operative) fields	Available confirmed via MQTT/ocpplib.log <i>vendorId/vendorErrorCode absent (optional per OCPP 1.6 4.7)*</i>	PASS
1.17	GetDiagnostics fileName	HTTP 200 <i>*discretionary (discretionary)</i>	PASS
1.18	DiagnosticsStatusNotification status	Not observed in ocpplib.log within 6 s <i>*discretionary (discretionary)</i>	WARN

Item	Test Description	Detail / Evidence	Result
1.19	UpdateFirmware	HTTP 200 <i>*discretionary (discretionary)</i>	PASS
1.20	FirmwareStatusNotification status	Not observed in ocpplib.log within 6 s <i>*discretionary (discretionary)</i>	WARN
1.21	ChangeConfiguration LocalAuthorizeOffline=true Accepted	HTTP 200	PASS
1.22	SendLocalList (Full, 1 entry) Accepted	HTTP 200	PASS
1.23	GetLocalListVersion listVersion	HTTP 200	PASS
1.24	ClearCache status Accepted	HTTP 200	PASS

Notes

Sandbox limitations (items 1.3–1.7)

The MEA sandbox REST API does not expose `/remote/triggerMessage` (HTTP 404). `TriggerMessage` requires the CSMS to initiate the command; items 1.4, 1.6, 1.8 depend on it.

ChangeAvailability (items 1.13–1.16)

`/remote/changeAvailability` returns HTTP 404. `ChangeAvailability` was issued via vSECC MQTT `set_availability`. The resulting `StatusNotification` frames were confirmed via `ocpplib.log`.

GetConfiguration (item 1.9)

MEA CSMS processes `GetConfiguration` asynchronously (HTTP 200, empty body). The vSECC accepts `ChangeConfiguration` for all standard keys (1.10–1.12, 1.21 PASS).

DiagnosticsStatus / FirmwareStatus (items 1.18, 1.20)

Discretionary items. vSECC does not send these notifications without a real FTP endpoint.

4 Section 2: Auto Charge Verification

3 PASS 0 FAIL 18 WARN 0 SKIP (21 total)

Item	Test Description	Detail / Evidence	Result
2.1	ChangeConfiguration AutoCharge=False Accepted	HTTP 200	PASS
""			
2.2	StatusNotification Preparing (AutoCharge=False)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
2.3	StatusNotification Available (unplug)	Depends on 2.2 (EV plug)	WARN
2.4	ChangeConfiguration AutoCharge=True Accepted	HTTP 200	PASS
""			
2.5	StatusNotification Preparing (AutoCharge=True)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
2.6	Authorize (VID) Accepted	Depends on 2.5 (EV plug)	WARN
2.7	StartTransaction (session started)	Depends on 2.5 (EV plug)	WARN
2.8	StatusNotification Charging	Depends on 2.5 (EV plug)	WARN
2.9	MeterValues (measurands during charging)	Depends on 2.5 (EV plug / active charging)	WARN
2.10	RemoteStopTransaction Accepted	HTTP 200	PASS
""			
2.11	StopTransaction (sent after RemoteStop)	session_state idle/finished not observed on MQTT in 30 s	WARN
2.12	StatusNotification Finishing	Finishing status not observed on MQTT <i>Some implementations go directly Available without Finishing</i>	WARN
2.13	StatusNotification Available (after RemoteStop/unplug)	Available status not observed on MQTT in 30 s	WARN
2.14	StatusNotification Preparing (Session 2)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
2.15	Authorize (Session 2) Accepted	Depends on 2.14 (EV plug)	WARN
2.16	StartTransaction (Session 2)	Depends on 2.14 (EV plug)	WARN
2.17	StatusNotification Charging (Session 2)	Depends on 2.14 (EV plug)	WARN
2.18	MeterValues (Session 2)	Depends on 2.14 (EV plug)	WARN
2.19	StatusNotification SuspendedEV	Depends on 2.14 (EV plug)	WARN
2.20	StopTransaction (reason=EVDisconnected)	Depends on 2.14 (EV plug)	WARN
2.21	StatusNotification Available (Session 2 end)	Depends on 2.14 (EV plug)	WARN

Notes

EV-dependent items (2.2, 2.5, 2.14 and dependents)

These items require a physical EV to be plugged into the vSECC connector. All items that depend on an active charging session (Authorize, StartTransaction, Charging, MeterValues, StopTransaction, etc.) are WARN when no EV is connected.

MeterValues (items 2.9, 2.18)

MEA sandbox does not expose TriggerMessage (HTTP 404). MeterValues are observable only if published autonomously on MeterValueSampleInterval.

StatusNotification SuspendedEV (item 2.19)

Requires the EV BMS to stop drawing current; cannot be reliably triggered without a real EV.

5 Section 3: Normal Charging Verification

3 PASS 0 FAIL 16 WARN 0 SKIP (19 total)

Item	Test Description	Detail / Evidence	Result
3.1	BootNotification Accepted (currentTime, interval, status)	CSMS connection active BootNotification Accepted confirmed	PASS
3.2	StatusNotification Available (initial state)	Available not observed on MQTT in 10 s <i>Check CSMS event log for StatusNotification Available on boot</i>	WARN
3.3	StatusNotification Preparing (EV plug, manual)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
3.4	Authorize (RFID card) Accepted	Depends on 3.3 (EV plug)	WARN
3.5	StartTransaction (manual, meterStart, idTag, connectorId)	Depends on 3.3 (EV plug)	WARN
3.6	StatusNotification Charging (manual session)	Depends on 3.3 (EV plug)	WARN
3.7	MeterValues (energy measurands, manual session)	Depends on 3.3 (EV plug)	WARN
3.8	StopTransaction (manual card tap, reason=Local)	Depends on 3.3 (EV plug)	WARN
3.9	StatusNotification Finishing (manual, EV still plugged)	Depends on 3.3 (EV plug)	WARN
3.10	StatusNotification Available (manual, EV unplug)	Depends on 3.3 (EV plug)	WARN
3.11	StatusNotification Preparing (EV plug, remote)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
3.12	RemoteStartTransaction Accepted	HTTP 200	PASS
""			
3.13	StartTransaction (triggered by RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
3.14	StatusNotification Charging (remote session)	Charging status not observed on MQTT in 20 s	WARN
3.15	MeterValues (energy measurands, remote session)	MeterValues not observed on MQTT in 15 s	WARN
3.16	RemoteStopTransaction Accepted	HTTP 200	PASS
""			
3.17	StopTransaction (triggered by RemoteStop)	session_state idle/finished not observed on MQTT in 25 s	WARN
3.18	StatusNotification Finishing (remote, EV still plugged)	Finishing not observed on MQTT in 15 s <i>Some implementations skip Finishing go directly to Available</i>	WARN
3.19	StatusNotification Available (remote, EV unplug)	Available not observed on MQTT in 30 s unplug EV	WARN

Notes

EV-dependent items

Items 3.3, 3.11 require a physical EV plug event. All downstream items are WARN without an EV.

RemoteStartTransaction (items 3.12–3.15)

RemoteStart is issued via MEA REST API. Session items confirmed via MQTT or `ocpplib.log`.

MeterValues (items 3.7, 3.15)

MEA sandbox does not expose TriggerMessage (HTTP 404). Values observed via MQTT autonomous publication or `ocpplib.log`.

6 Section 4: Reset Verification

6 PASS 0 FAIL 15 WARN 0 SKIP (21 total)

Item	Test Description	Detail / Evidence	Result
4.1	Reset (Hard) Accepted	HTTP 200	PASS
""			
4.2	BootNotification Accepted (after Hard Reset)	CSMS reconnection confirmed after reset	PASS
4.3	StatusNotification Available (after reboot)	Available not observed on MQTT in 20 s after reset	WARN
4.4	StatusNotification Preparing (EV plug, Flow 1)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
4.5	RemoteStartTransaction Accepted	HTTP 200	PASS
""			
4.6	StartTransaction (triggered by RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
4.7	StatusNotification Charging (Flow 1)	Charging status not observed on MQTT in 20 s	WARN
4.8	MeterValues (Flow 1, during charging)	MeterValues not observed on MQTT in 15 s	WARN
4.9	Reset (Soft) Accepted (during charging)	HTTP 200	PASS
""			
4.10	StopTransaction (triggered by Soft Reset)	session_state idle/finished not observed in 30 s <i>Soft Reset should cause graceful StopTransaction before reboot</i>	WARN
4.11	StatusNotification Finishing (Soft Reset, EV plugged)	Finishing not observed on MQTT in 15 s <i>Some implementations skip Finishing go directly to Available</i>	WARN
4.12	StatusNotification Available (unplug after Soft Reset)	Available not observed on MQTT in 30 s unplug EV	WARN
4.13	StatusNotification Preparing (EV plug, Flow 2)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
4.14	RemoteStartTransaction Accepted (Flow 2)	HTTP 200	PASS
""			
4.15	StartTransaction (Flow 2, triggered by RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
4.16	StatusNotification Charging (Flow 2)	Charging status not observed on MQTT in 20 s	WARN
4.17	MeterValues (Flow 2, during charging)	MeterValues not observed on MQTT in 15 s	WARN
4.18	Reset (Hard) Accepted (during charging)	HTTP 200	PASS
""			
4.19	StopTransaction (triggered by Hard Reset)	session_state idle/finished not observed in 30 s <i>Hard Reset may not send StopTransaction before rebooting</i>	WARN

Item	Test Description	Detail / Evidence	Result
4.20	StatusNotification Finishing (Hard Reset, EV plugged)	Finishing not observed on MQTT in 15 s <i>Hard Reset typically skips Finishing goes directly Available after reboot</i>	WARN
4.21	StatusNotification Available (after Hard Reset, EV unplug)	Available not observed in 90s unplug EV / wait for reboot	WARN

Notes

Reset flows

Hard Reset and Soft Reset commands are issued via MEA REST API. BootNotification after Hard Reset confirms vSECC reconnection. All dependent session items (StartTransaction, MeterValues, StopTransaction) require a physical EV to be connected during the test.

BootNotification after reset

Verified via MQTT `vsecc/ocpp_connection_status` reconnect event. The `ocpplib.log` provides additional frame-level evidence.

7 Section 5: Reservation Verification

6 PASS 0 FAIL 13 WARN 0 SKIP (19 total)

Item	Test Description	Detail / Evidence	Result
5.1	StatusNotification Available (initial)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
5.2	ReserveNow (5 min) Accepted	HTTP 200 <i>reservation_id not found in response</i>	PASS
"			
5.3	StatusNotification Reserved (after ReserveNow)	Reserved status not observed on MQTT in 15 s <i>vSECC may use different status reporting for reservations</i>	WARN
5.4	CancelReservation (id=0) Accepted	HTTP 200	PASS
"			
5.5	StatusNotification Available (after cancel)	Available not observed on MQTT in 15 s after cancel	WARN
5.6	ReserveNow (short, for expiry test) Accepted	HTTP 200 <i>reservation_id not found</i>	PASS
"			
5.7	Reservation expiry (wait 90s)	Available not observed in 90s <i>MEA CSMS may enforce a minimum reservation duration; expiry auto-cancelled</i>	WARN
5.8	StatusNotification Available (after expiry)	Available not observed on MQTT in 15 s after expiry/cancel	WARN
5.9	ReserveNow (10 min, for charge session) Accepted	HTTP 200 <i>reservation_id not found</i>	PASS
"			
5.10	StatusNotification Reserved (Flow 3)	Reserved status not observed on MQTT in 15 s	WARN
5.11	StatusNotification Reserved/Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
5.12	RemoteStartTransaction Accepted	HTTP 200	PASS
"			
5.13	StartTransaction (triggered by RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
5.14	StatusNotification Charging	Charging status not observed on MQTT in 20 s	WARN
5.15	MeterValues (during reservation charge)	MeterValues not observed on MQTT in 15 s	WARN
5.16	RemoteStopTransaction Accepted	HTTP 200	PASS
"			
5.17	StopTransaction (triggered by RemoteStop)	session_state idle/finished not observed on MQTT in 25 s	WARN
5.18	StatusNotification Finishing (EV still plugged)	Finishing not observed on MQTT in 15 s <i>Some implementations skip Finishing go directly to Available</i>	WARN

Item	Test Description	Detail / Evidence	Result
5.19	StatusNotification Available (EV unplug)	Available not observed on MQTT in 30 s unplug EV	WARN

Notes

Reservation flow

ReserveNow and CancelReservation are issued via MEA REST API. StatusNotification Reserved/Available transitions are confirmed via MQTT or `ocpplib.log`.

EV-dependent items

Items 5.11 (EV plug) and all downstream charging session items require a physical EV.

Reservation expiry (item 5.7)

The vSECC must release the reservation automatically when the timer expires. Observed via MQTT StatusNotification Available after the expiry window.

8 Section 6: Charging Profile Verification

4 PASS 0 FAIL 22 WARN 0 SKIP (26 total)

Item	Test Description	Detail / Evidence	Result
6.1	StatusNotification Available (initial state)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
6.2	StatusNotification Preparing (EV plug, Flow 1)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
6.3	Authorize (RFID card) Accepted	Depends on 6.2 (EV plug)	WARN
6.4	StartTransaction (manual RFID, meterStart, idTag, connectorId)	Depends on 6.2 (EV plug)	WARN
6.5	StatusNotification Charging (Flow 1)	Depends on 6.2 (EV plug)	WARN
6.6	MeterValues (Flow 1, before SetChargingProfile)	Depends on 6.2 (EV plug)	WARN
6.7	SetChargingProfile (TxProfile, 5 kW, stackLevel=1) Accepted	Depends on 6.2 (EV plug)	WARN
6.8	MeterValues (Flow 1, after first SetChargingProfile)	Depends on 6.2 (EV plug)	WARN
6.9	SetChargingProfile (update, 7.4 kW, stackLevel=1) Accepted	Depends on 6.2 (EV plug)	WARN
6.10	MeterValues (Flow 1, after second SetChargingProfile)	Depends on 6.2 (EV plug)	WARN
6.11	StopTransaction (manual card tap, reason=Local)	Depends on 6.2 (EV plug)	WARN
6.12	StatusNotification Finishing (Flow 1, EV still plugged)	Depends on 6.2 (EV plug)	WARN
6.13	StatusNotification Available (Flow 1, EV unplug)	Depends on 6.2 (EV plug)	WARN
6.14	StatusNotification Preparing (EV plug, Flow 2)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
6.15	RemoteStartTransaction Accepted	HTTP 200	PASS
""			
6.16	StartTransaction (triggered by RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
6.17	StatusNotification Charging (Flow 2)	Charging status not observed on MQTT in 20 s	WARN
6.18	MeterValues (Flow 2, before SetChargingProfile)	MeterValues not observed on MQTT in 15 s	WARN
6.19	SetChargingProfile (Flow 2, TxProfile 5 kW) Accepted	HTTP 200	PASS
""			
6.20	MeterValues (Flow 2, after first SetChargingProfile)	MeterValues not observed on MQTT in 15 s	WARN
6.21	SetChargingProfile (Flow 2, update 7.4 kW) Accepted	HTTP 200	PASS
""			
6.22	MeterValues (Flow 2, after second SetChargingProfile)	MeterValues not observed on MQTT in 15 s	WARN

Item	Test Description	Detail / Evidence	Result
6.23	RemoteStopTransaction Accepted	HTTP 200	PASS
"			
6.24	StopTransaction (triggered by RemoteStop)	session_state idle/finished not observed on MQTT in 25 s	WARN
6.25	StatusNotification Finishing (Flow 2, EV still plugged)	Finishing not observed on MQTT in 15 s <i>Some implementations skip Finishing go directly to Available</i>	WARN
6.26	StatusNotification Available (Flow 2, EV unplug)	Available not observed on MQTT in 30 s unplug EV	WARN

Notes

Charging profile flows

SetChargingProfile (TxProfile) commands are issued via MEA REST API during an active session. MeterValues before/after profile changes confirm the power limit adjustment.

EV-dependent items

All charging session items require a physical EV. Without an EV the test records WARN.

9 Section 7: Abnormal Operation Verification

9 PASS 0 FAIL 45 WARN 0 SKIP (54 total)

Item	Test Description	Detail / Evidence	Result
7.1.1	StatusNotification Available (charger idle, no EV)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.1.2	RemoteStartTransaction (EV not connected) HTTP 200	HTTP 200 <i>vSECC should respond Rejected when no EV present</i>	PASS
""			
7.2.1	StatusNotification Available (before EV plug)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.2.2	StatusNotification Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
7.2.3	RemoteStartTransaction (first) HTTP 200	HTTP 200	PASS
""			
7.2.4	RemoteStartTransaction (second, concurrent) HTTP 200	HTTP 200 <i>CSMS likely blocks duplicate; charger-level Rejected not observable without proxy</i>	PASS
""			
7.2.5	StartTransaction (session started after first RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
7.2.6	StatusNotification Charging	Charging status not observed on MQTT in 20 s	WARN
7.2.7	MeterValues (during charging)	MeterValues not observed on MQTT in 15 s	WARN
7.2.8	RemoteStopTransaction HTTP 200	HTTP 200	PASS
""			
7.2.9	StopTransaction (session ended after RemoteStop)	session_state idle/finished not observed on MQTT in 25 s	WARN
7.2.10	StatusNotification Finishing	Finishing not observed on MQTT in 15 s <i>Some implementations skip Finishing and go directly to Available</i>	WARN
7.2.11	StatusNotification Available (after stop / unplug)	Available not observed on MQTT in 30 s unplug EV	WARN
7.3.1	StatusNotification Available (before EV plug)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.3.2	StatusNotification Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
7.3.3	Authorize (invalid RFID tag) Rejected	Not triggerable via API vSECC reads RFID directly from hardware <i>Requires physical RFID scan on vSECC hardware of unknown/blocked card</i>	WARN
7.3.4	Authorize (valid RFID tag) Accepted	Not triggerable via API vSECC reads RFID directly from hardware <i>Requires physical RFID scan on vSECC hardware of authorized card after rejected card</i>	WARN

Item	Test Description	Detail / Evidence	Result
7.3.5	StartTransaction (after valid RFID)	Depends on 7.3.3/7.3.4 (physical RFID scan) <i>Session starts only after valid RFID tap on vSECC hardware</i>	WARN
7.3.6	StatusNotification Charging (after valid auth)	Depends on 7.3.5 (StartTransaction) <i>Requires completed Authorize sequence via physical RFID hardware</i>	WARN
7.3.7	MeterValues (during charging)	Depends on 7.3.6 (Charging state) <i>Requires completed Authorize sequence via physical RFID hardware</i>	WARN
7.3.8	RemoteStopTransaction	Depends on 7.3.5 (StartTransaction) <i>Requires active session started via physical RFID hardware</i>	WARN
7.3.9	StopTransaction (after RemoteStop)	Depends on 7.3.8 (RemoteStop) <i>Requires active session started via physical RFID hardware</i>	WARN
7.3.10	StatusNotification Finishing	Depends on 7.3.8 (RemoteStop) <i>Requires active session started via physical RFID hardware</i>	WARN
7.3.11	StatusNotification Available (after stop / unplug)	Depends on 7.3.8 (RemoteStop) <i>Requires active session started via physical RFID hardware</i>	WARN
7.4.1	StatusNotification Available (before EV plug)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.4.2	StatusNotification Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
7.4.3	RemoteStartTransaction HTTP 200	HTTP 200	PASS
""			
7.4.4	StartTransaction (session started after RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
7.4.5	StatusNotification Charging	Charging status not observed on MQTT in 20 s	WARN
7.4.6	MeterValues (during charging)	MeterValues not observed on MQTT in 15 s	WARN
7.4.7	StopTransaction (reason=EmergencyStop)	Triggered by physical emergency stop button on vSECC hardware <i>Requires physical emergency stop button press on the vSECC unit</i>	WARN
7.4.8	StatusNotification Faulted (after EmergencyStop)	Faulted status not observed on MQTT (E-stop not pressed) <i>Requires physical emergency stop button press on the vSECC unit</i>	WARN
7.4.9	StatusNotification Available (E-stop cleared)	Depends on 7.4.7 (physical E-stop press and release) <i>E-stop must be physically released to restore normal operation</i>	WARN
7.4.10	StatusNotification Available (EV unplug after E-stop recovery)	Depends on 7.4.7 (physical E-stop press and release) <i>EV must be unplugged after E-stop recovery to reach Available</i>	WARN

Item	Test Description	Detail / Evidence	Result
7.5.1	StatusNotification Available (before door open)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.5.2	StatusNotification Faulted (reason=DoorOpen)	Triggered by opening physical door/panel on vSECC hardware <i>Requires physical door open on vSECC hardware; not triggerable via API</i>	WARN
7.5.3	StatusNotification Available (after door closed)	Depends on 7.5.2 (physical door open and close) <i>Door must be physically closed to restore normal operation</i>	WARN
7.6.1	StatusNotification Available (before EV plug)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.6.2	StatusNotification Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
7.6.3	RemoteStartTransaction HTTP 200	HTTP 200	PASS
""			
7.6.4	StartTransaction (session started after RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
7.6.5	StatusNotification Charging	Charging status not observed on MQTT in 20 s	WARN
7.6.6	MeterValues (during charging)	MeterValues not observed on MQTT in 15 s	WARN
7.6.7	StopTransaction (PowerLoss) simulated via Hard Reset HTTP 200	HTTP 200 <i>Simulated via Hard Reset; physical power loss not testable in this environment</i>	PASS
""			
7.6.8	BootNotification Accepted (vSECC reconnected after reboot)	CSMS reconnection confirmed after Hard Reset	PASS
7.6.9	StatusNotification Preparing (EV still connected after reboot)	Preparing not observed on MQTT in 20 s after reboot <i>EV must remain connected through the reboot sequence</i>	WARN
7.6.10	StatusNotification Available (EV unplug after reboot)	Available not observed on MQTT in 30 s unplug EV	WARN
7.7.1	SendLocalList (version=1, RFID_TEST=Accepted) HTTP 200	HTTP 200	PASS
""			
7.7.2	StatusNotification Available (charger online before offline test)	Offline sequence not observable via MQTT/REST without proxy <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.3	StatusNotification Preparing (EV plug while charger offline)	Offline start requires disconnecting vSECC from CSMS network <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.4	Authorize (local list lookup while offline) Accepted	vSECC must authorize from local list while CSMS unreachable <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN

Item	Test Description	Detail / Evidence	Result
7.7.5	StartTransaction (offline, buffered by vSECC)	vSECC buffers transaction in firmware; not observable without proxy <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.6	StopTransaction (offline, buffered by vSECC)	vSECC buffers stop in firmware; delivered when CSMS reconnects <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.7	CSMS reconnect buffered Start/StopTransaction flushed	Reconnect flushing is firmware-internal; not observable without proxy <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN

Notes

Remote-controllable flows (7.2, 7.6)

RemoteStart/RemoteStop and Reset commands are verifiable via MEA REST API and `ocpplib.log`.

Physical-hardware items (7.3, 7.4.7–7.4.10, 7.5)

Card-swap, emergency stop, door-open switch, and power-loss scenarios require physical interaction with the hardware. These are permanently recorded as WARN with a remark indicating the required operator action.

Offline buffering (7.7)

Offline transaction buffering cannot be verified without direct OCPP frame capture from the reconnection burst. Recorded as WARN pending further tooling.

10 Section 8: Dual Connector Verification

0 PASS 0 FAIL 0 WARN 50 SKIP (50 total)

Item	Test Description	Detail / Evidence	Result
8.1.1	Concurrent RemoteStart Connector 1 Accepted	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.2	Concurrent RemoteStart Connector 2 Accepted	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.3	StartTransaction Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.4	StartTransaction Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.5	StatusNotification Charging Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.6	StatusNotification Charging Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.7	MeterValues Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.8	MeterValues Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.9	RemoteStop Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.10	RemoteStop Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.11	StopTransaction Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.12	StopTransaction Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.13	StatusNotification Finishing Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.14	StatusNotification Finishing Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.15	StatusNotification Available Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.16	StatusNotification Available Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.17	Transaction isolation Connector 1 not affected by Connector 2 stop	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.1.18	Transaction isolation Connector 2 not affected by Connector 1 stop	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.1	Emergency Stop signal sent	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.2	RemoteStop broadcast Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.3	RemoteStop broadcast Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.4	StopTransaction (EmergencyStop reason) Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.5	StopTransaction (EmergencyStop reason) Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP

Item	Test Description	Detail / Evidence	Result
8.2.6	StatusNotification Faulted Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.7	StatusNotification Faulted Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.8	StatusNotification Unavailable (after fault clear) Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.9	StatusNotification Unavailable (after fault clear) Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.10	ChangeAvailability Operative Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.11	ChangeAvailability Operative Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.12	StatusNotification Available Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.13	StatusNotification Available Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.14	New session possible after emergency stop Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.15	New session possible after emergency stop Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.2.16	Shared emergency stop clears both connectors simultaneously	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.1	Power loss event detected	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.2	Active session aborted Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.3	Active session aborted Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.4	StopTransaction (PowerLoss reason) Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.5	StopTransaction (PowerLoss reason) Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.6	vSECC power loss BootNotification after restore	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.7	StatusNotification Available after power restore Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.8	StatusNotification Available after power restore Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.9	Session data integrity after power restore Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.10	Session data integrity after power restore Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.11	RemoteStart accepted after power restore Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.12	RemoteStart accepted after power restore Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.13	StartTransaction after power restore Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP

Item	Test Description	Detail / Evidence	Result
8.3.14	StartTransaction after power restore Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.15	StatusNotification Charging after power restore Connector 1	vSECC.single has one connector dual-connector scenarios N/A	SKIP
8.3.16	StatusNotification Charging after power restore Connector 2	vSECC.single has one connector dual-connector scenarios N/A	SKIP

Notes

Dual connector

The Vector vSECC.single Board supports only a single connector. All Section 8 dual-connector items are recorded as SKIP.

11 Section 9: MEA-Specific Configuration & V2G/BPT

5 PASS 0 FAIL 3 WARN 0 SKIP (8 total)

Item	Test Description	Detail / Evidence	Result
9.0	CSMS connection probe GetConfiguration HTTP 200	vSECC connected to MEA CSMS	PASS
9.1	ChangeConfiguration V2GMode=true Accepted	HTTP 200	PASS
"			
9.2	ChangeConfiguration MeterValueSampleInterval=10 Accepted	HTTP 200	PASS
"			
9.3	ChangeConfiguration LocalAuthorizeOffline=false Accepted	HTTP 200	PASS
"			
9.4.0	ChangeConfiguration V2GMode=true (pre-condition for power demand)	HTTP 200	PASS
"			
9.4.1	Power Demand -5000 W (BPT discharge from EV to grid)	HTTP 200; <i>metervalues not observed in 15s</i> <i>V2G/BPT requires active charging session and</i> <i>V2G-capable EV</i>	WARN
"			
9.4.2	Power Demand +2000 W (BPT charge EV from grid)	HTTP 200; <i>metervalues not observed in 15s</i> <i>V2G/BPT requires active charging session and</i> <i>V2G-capable EV</i>	WARN
"			
9.4.3	Power Demand 0 W (BPT idle / stop power flow)	HTTP 200; <i>metervalues not observed in 15s</i> <i>V2G/BPT requires active charging session and</i> <i>V2G-capable EV</i>	WARN
"			

Notes

V2G/BPT configuration (items 9.1–9.4)

ChangeConfiguration commands (V2GMode, MeterValueSampleInterval, LocalAuthorizeOffline) are issued via MEA REST API and return HTTP 200.

V2G power demand (items 9.4.1–9.4.3)

Require an active ISO 15118 V2G charging session (physical EV with V2G capability). Observed via MQTT MeterValues with V2G measurands or `ocpplib.log`.

12 Section 10: CSMS Command Verification

14 PASS 0 FAIL 9 WARN 0 SKIP (23 total)

Item	Test Description	Detail / Evidence	Result
10.1	Authorize	Authorize is CSCSMS; not triggerable via REST API <i>Observed via MQTT during charging sessions (charging_authorization_state)</i>	WARN
10.2	BootNotification	BootNotification is CSCSMS; confirmed via CSMS connection (Section 1 item 1.1) <i>Charger sends BootNotification autonomously on connect/reboot</i>	WARN
10.3	Data Transfer vendorId=MEA messageId=Test	HTTP 200 status=Bad Request	PASS
{ "status": "Bad Request", "code": 400, "messages": "Additional properties are not allowed ('chargepoint_id', 'messageId', ' "			
10.4	Heartbeat	HTTP 404 <i>Heartbeat endpoint not exposed by MEA sandbox; charger sends heartbeats autonomously</i>	WARN
{ "detail": "Not Found" }			
10.5	MeterValues	MeterValues is CSCSMS; TriggerMessage not supported (HTTP 404) <i>Observable on MQTT metervalues topic during active charging session</i>	WARN
10.6	RemoteStartTransaction Accepted	HTTP 200	PASS
"			
10.7	RemoteStopTransaction (transaction_id=0) Accepted	HTTP 200	PASS
"			
10.8	Reset (Hard) Accepted	HTTP 200	PASS
"			
10.9	StartTransaction	CSCSMS; observable via MQTT charging_session_state during charging session <i>Charger sends StartTransaction automatically when session begins</i>	WARN
10.10	StatusNotification	CSCSMS; observable via MQTT vsec/connector/+/status/# <i>Charger sends StatusNotification on connector state changes</i>	WARN
10.11	StopTransaction	CSCSMS; observable via MQTT charging_session_state <i>Charger sends StopTransaction on session end (RemoteStop / EVDisconnect / Reset)</i>	WARN
10.12	ReserveNow (connector_id=1, reservationId=1) Accepted	HTTP 200	PASS
"			
10.13	CancelReservation (reservationId=1) Accepted	HTTP 200	PASS
"			
10.14	SetChargingProfile (TxDefaultProfile, 32 A) Accepted	HTTP 200	PASS

Item	Test Description	Detail / Evidence	Result
"			
10.15	TriggerMessage (StatusNotification)	MEA sandbox /remote/triggerMessage not exposed (HTTP 404) <i>CSMSCS command; not testable via MEA REST API</i>	WARN
10.16	GetConfiguration Accepted	HTTP 200	PASS
"			
10.17	ChangeConfiguration HeartbeatInterval=600 Accepted	HTTP 200	PASS
"			
10.18	ChangeAvailability (connector 1, Inoperative)	Unavailable not observed on MQTT in 10 s after MQTT set_availability <i>MEA REST /remote/changeAvailability 404; MQTT fallback used</i>	WARN
10.19	GetDiagnostics (ftp://example.com)	HTTP 200	PASS
"			
10.20	UpdateFirmware (ftp://example.com)	HTTP 200	PASS
"			
10.21	SendLocalList (Full, version=1, idTag=RFID_TEST) Accepted	HTTP 200	PASS
"			
10.22	GetLocalListVersion Accepted	HTTP 200	PASS
"			
10.23	ClearCache Accepted	HTTP 200	PASS
"			

Notes

CS→CSMS items (10.1, 10.2, 10.5, 10.9–10.11)

These OCPP actions are charger-initiated and cannot be triggered via the MEA REST API. The test checks `ocpplib.log` for recent »> frames as secondary evidence. A PASS requires the frame to appear in the log within a 3 s window.

TriggerMessage (item 10.15)

The MEA sandbox does not expose `/remote/triggerMessage` (HTTP 404).

ChangeAvailability (item 10.18)

`/remote/changeAvailability` returns HTTP 404. Verified via MQTT `set_availability` with `ocpplib.log` fallback.

GetDiagnostics / UpdateFirmware (items 10.19–10.20)

MEA sandbox returns HTTP 404 for these endpoints (PascalCase path required but both 404).

13 Section 11: Performance – Reconnect Time

2 PASS 0 FAIL 0 WARN 0 SKIP (2 total)

Item	Test Description	Detail / Evidence	Result
11.1	Reconnect Time after Hard Reset	Reconnected in 0.1 s (< 90 s threshold) <i>time_ms=146</i>	PASS
"			
11.2	Average Reconnect Time (3 measurements)	avg=0.2s min=0.2s max=0.2s [0.2, 0.2, 0.2] <i>run1=247ms; run2=156ms; run3=161ms</i>	PASS
"			

Notes

Reconnect time measurement

The vSECC is hard-reset via MEA REST API and the test measures elapsed time until `vsecc/ocpp_connection_status = connected` is observed on MQTT. PASS threshold: < 90s. WARN: 90–180s. FAIL: > 180s or no reconnect. Three measurements are taken for the average (item 11.2). `ocpplib.log` provides BootNotification frame timestamps as supplementary evidence.